

Buddy AI - Hardware Pin Map

Locked stack: Raspberry Pi 3 B+ (brain + Pi Camera), ESP32-WROOM (motors, pan-tilt servos, distance sensor), Bluetooth speaker (voice out), phone/laptop mic for v1 (no onboard mic yet). ESP32-CAM is spare / not required.

1. Power

From	To	Notes
Pi 5V PSU / power bank	Pi 3 B+ power input	Do not power motors from the Pi
Motor battery (+)	Motor driver VS / VCC motor	Match motor voltage (often 6-12V)
Motor battery (-)	Motor driver GND	
Regulated 5V	ESP32-WROOM VIN / 5V	Buck from motor pack if needed
ESP32 GND	Driver GND	Must share ground with driver
Servo 5V supply	Servo red wires	Do NOT use ESP32 3.3V for servos
Servo GND	ESP32 GND	Common ground
Ultrasonic VCC	5V	HC-SR04 prefers 5V
Ultrasonic GND	ESP32 GND	Common ground

2. Pi Camera -> Raspberry Pi 3 B+

Use the CSI ribbon cable into the CSI connector (between HDMI and audio). No GPIO pins. Enable camera in raspi-config / use libcamera. Orient the ribbon per the official Pi Camera guide (contacts toward HDMI on most Pi 3 setups).

3. ESP32-WROOM -> Motor driver (firmware defaults)

Left motor = A, Right motor = B. Pins match esp32-motor/config.h.

ESP32 GPIO	L298N	L9110	Function
GPIO 26	IN1	A-IA	Left / Motor A dir 1
GPIO 27	IN2	A-IB	Left / Motor A dir 2
GPIO 25	ENA (PWM)	(unused*)	Left speed PWM
GPIO 33	IN3	B-IA	Right / Motor B dir 1
GPIO 32	IN4	B-IB	Right / Motor B dir 2
GPIO 14	ENB (PWM)	(unused*)	Right speed PWM
GND	GND	GND	Common ground

* For L9110 / drivers without ENA/ENB: set MOTOR_HAS_PWM to 0 in config.h. Driver OUT1/OUT2 -> left motor; OUT3/OUT4 -> right motor. If a side runs backward, swap that motor's two wires or set MOTOR_INVERT_LEFT/RIGHT.

4. Pan-tilt servos -> ESP32-WROOM

Servo wire	Connect to
Brown / Black (GND)	ESP32 GND
Red (VCC)	External 5V

Buddy AI Finder Robot - Pin Connections

Servo wire	Connect to
Orange / Yellow signal - Pan	GPIO 18
Orange / Yellow signal - Tilt	GPIO 19

5. Distance sensor -> ESP32-WROOM

HC-SR04 ultrasonic (recommended defaults):

HC-SR04	ESP32
VCC	5V
GND	GND
TRIG	GPIO 12
ECHO	GPIO 13 (use resistor divider 5V->3.3V if possible)

If using VL53L0X ToF instead:

VL53L0X	ESP32
VCC	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

6. Audio / network (no GPIO)

Device	Connection
Bluetooth speaker	Pair in Raspberry Pi OS (voice out)
Phone / laptop mic (v1)	Wi-Fi to Pi web UI / POST /command
ReSpeaker 2-Mic HAT (next version)	Onboard voice-in - not wired in v1
ESP32-CAM	Spare - not required when using Pi Camera

7. ESP32-WROOM pin summary

GPIO	Function
26	Motor A IN1
27	Motor A IN2
25	Motor A ENA (PWM)
33	Motor B IN1
32	Motor B IN2
14	Motor B ENB (PWM)
18	Pan servo signal
19	Tilt servo signal
12	Ultrasonic TRIG
13	Ultrasonic ECHO
21 / 22	ToF SDA / SCL (if VL53 instead of HC-SR04)
GND	Common with driver, servos, sensor

8. Safety notes

- Never feed motor current through the Pi 5V rail.
- Share GND between ESP32, motor driver logic, servos, and sensors.
- Level-shift HC-SR04 ECHO (5V) down to 3.3V for ESP32 safety when possible.
- Servos need a solid 5V supply; brownouts cause random resets.
- Put Pi 3 B+ and ESP32-WROOM on the same Wi-Fi network.

9. Firmware / config references

Motor pins: `esp32-motor/config.h`

Wiring notes: `docs/wiring.md`

Repo: <https://github.com/AnuragShirodkar/Buddy-AI>